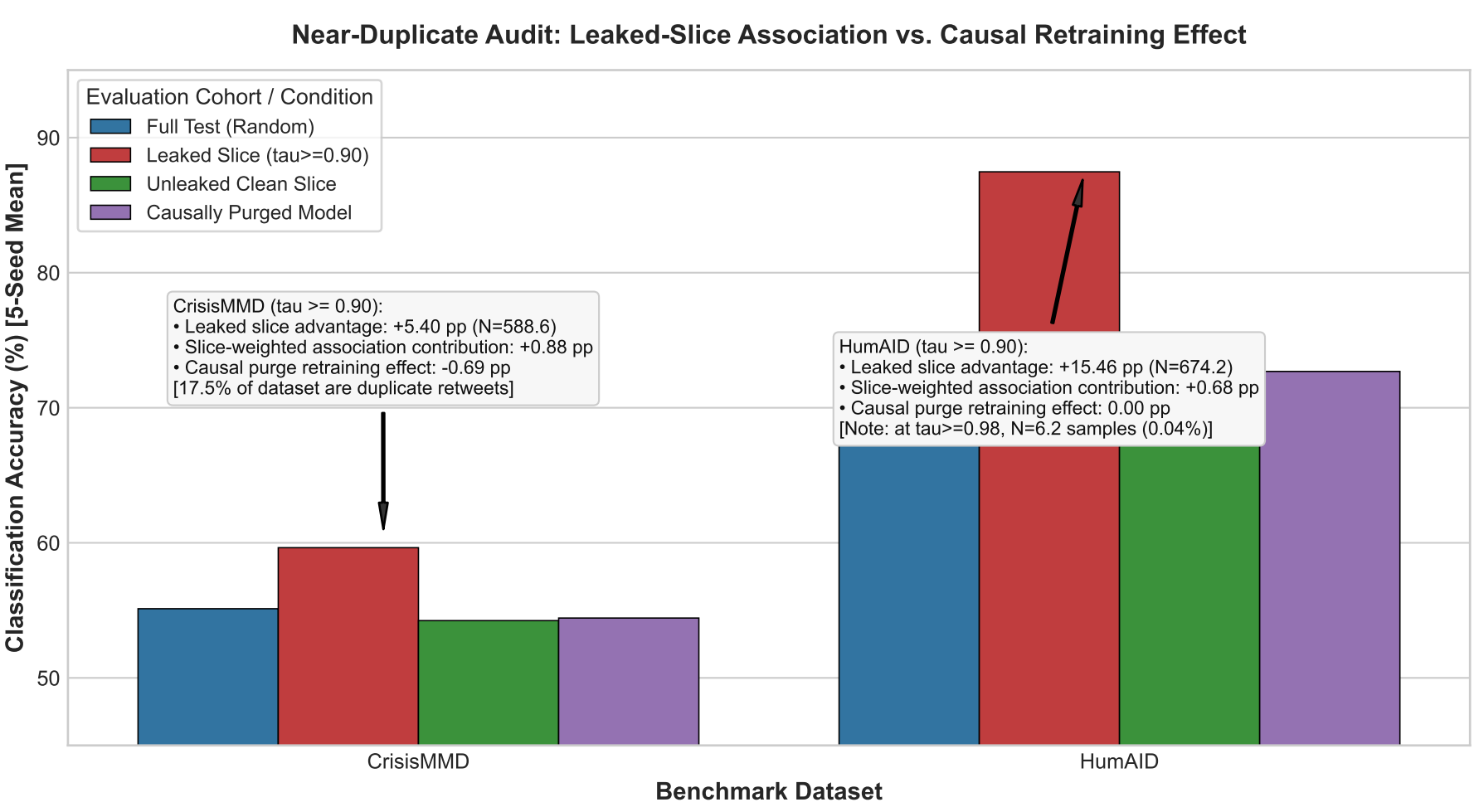
Near-Duplicate Audit: Leaked-Slice Association vs. Causal Retraining Effect
Evaluation Cohort / Condition
Full Test (Random)
Leaked Slice (tau>=0.90)
Unleaked Clean Slice
Causally Purged Model
CrisisMMD (tau >= 0.90):
• Leaked slice advantage: +5.40 pp (N=588.6)
• Slice-weighted association contribution: +0.88 pp
• Causal purge retraining effect: -0.69 pp
[17.5% of dataset are duplicate retweets]
HumAID (tau >= 0.90):
• Leaked slice advantage: +15.46 pp (N=674.2)
• Slice-weighted association contribution: +0.68 pp
• Causal purge retraining effect: 0.00 pp
[Note: at tau>=0.98, N=6.2 samples (0.04%)]
Classification Accuracy (%) [5-Seed Mean]
90
80
70
60
50
CrisisMMD
HumAID
Benchmark Dataset